

Topic 1.11 - Equivalent Representations of Polynomial and Rational Expressions

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Rewrite polynomial expressions in factored, expanded, and quotient-remainder forms and explain what each form reveals.
- Use polynomial division and the Remainder and Factor Theorems to make and justify conclusions.
- Rewrite rational expressions while preserving exclusions, holes, zeros, and asymptotes.

Section A: Foundation and representation

Question 1

Factoring

No calculator

Factor $x^3 + 2x^2 - 9x - 18$ completely and identify each zero.

Question 2

Expansion

No calculator

Expand $(x - 4)^2(x + 1)$ and state the degree, leading coefficient, and y -intercept.

Section A continued

Question 3

Representation

No calculator

For $P(x) = -2(x + 1)^3(x - 5)^2$, state two features visible immediately and one feature not immediately visible.

Question 4

Division

No calculator

Divide $3x^3 + x^2 - 10x + 8$ by $x + 2$ and state the quotient-remainder identity.

Question 5

Parameter

No calculator

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 9x - 9$.

Section B: Application and reasoning**Question 6**

Structure

No calculator

A polynomial satisfies $P(x) = (x + 4)(2x^2 - x + 3) - 11$. Find the quotient and remainder when divided by $x + 4$, and evaluate $P(-4)$.

Question 7

Rational structure

No calculator

Simplify $\frac{x^2+x-6}{x^2-4x+3}$ and identify all exclusions, holes, zeros, and vertical asymptotes.

Question 8

Error analysis

No calculator

A student writes $\frac{x^2-9}{x-3} = x + 3$ for all real x . Diagnose and repair the statement.

Section C: Challenge and error analysis

Question 9

Precision

No calculator

Do $f(x) = \frac{x^2-1}{x-1}$ and $g(x) = x + 1$ define the same function if no domain is specified? Justify.

Question 10

Challenge

No calculator

Expand $(3x + 2)^4$ using the Binomial Theorem.

Question 11

Challenge

No calculator

Find a and b so that $x^2 - 1$ is a factor of $P(x) = x^4 + ax^3 + bx^2 + ax + 1$.

Question 12

Modeling

No calculator

A model is written both as $A(t) = 120t - 300 + \frac{1500}{t+5}$ and $A(t) = \frac{120t^2 + 300t}{t+5}$. Explain which form better reveals long-run behavior and which better reveals the original denominator restriction.

Section D: AP-style multiple choice

MCQ 1

No calculator

What is the remainder when $P(x) = 2x^3 - x + 5$ is divided by $x - 2$?

- A. 3
- B. 11
- C. 19
- D. 23

MCQ 2

No calculator

Which expression is equivalent to $\frac{x^2-4}{x^2-x-2}$ on its original domain?

- A. $\frac{x+2}{x+1}, x \neq 2, -1$
- B. $\frac{x+2}{x+1}, x \neq -2, 1$
- C. $\frac{x-2}{x-1}$
- D. 1

MCQ 3

No calculator

If $P(c) = 0$, which statement must be true?

- A. $x + c$ is a factor
- B. $x - c$ is a factor
- C. c is the leading coefficient
- D. The degree is c

MCQ 4

No calculator

Which form most directly displays a polynomial zero of multiplicity 4 at $x = 3$?

- A. Expanded form
- B. Table form
- C. A factor $(x - 3)^4$
- D. Quotient-remainder form

AP-style free response 1**No calculator - 6 points**

Let $P(x) = x^4 - 5x^3 + 2x^2 + 20x - 24$.

A. Show that $x - 3$ is a factor and divide P by $x - 3$.

B. Factor P completely over the reals.

C. Describe every real zero and its crossing or touching behavior.

D. Simplify $\frac{P(x)}{(x-2)(x+2)}$, preserving restrictions and identifying holes.

AP-style free response 2

Calculator permitted - 6 points

A production model is $A(t) = \frac{120t^2 + 300t}{t+5}$ for $0 \leq t \leq 40$, where A is measured in units per day.

- A. Use polynomial division to write A in quotient-remainder form.

- B. Explain how the new form describes the model's long-run trend within its contextual domain.

- C. Solve $A(t) = 2000$ and report the relevant solution to three decimals.

- D. Explain why $t = -5$ remains important even though it lies outside the contextual domain.
